PENCIL / PAPER / SLIDE RULE

APOLLO

Mission Laboratory



Save the mission using math and your slipstick!

Two missions. Every command earns its GO.

Apollo 11 — Beginner

Land Eagle, conduct an EVA, rendezvous, and return.

Apollo 13 — Advanced

Turn Aquarius into a lifeboat and bring the crew home.

A cooperative, calculation-driven role-playing game
for one referee and one to three flight crew, or solo play.

20 mission checkpoints / printable working space / no dice

Companion: `apollo-answer-key.pdf`
Contains every calculation outline, worked answer, and referee ruling.

Inspired by NASA flight planning. Historical schedules and invented training models are distinguished throughout. An educational game, not flight software or an operational checklist.

Start here: a five-minute crew briefing

Equipment and session length

Print the ten cards for your mission and its ledger. Bring pencils, erasers, scrap paper, and at least one physical slide rule with C/D and A scales. An L scale is required for Apollo 13's PC+2 problem; a referee may supply a printed logarithmic scale if the rule lacks L. S/T scales are optional because a small tangent table is supplied. No electronic calculators during play. Allow 90–150 minutes for Apollo 11, or 2–3 hours for Apollo 13.

Choose a crew. For Apollo 11, play Neil Armstrong (commander, landing and decisions), Buzz Aldrin (LM pilot, budgets and burn pads), and Michael Collins (CM pilot, timing and rendezvous). For Apollo 13, play Jim Lovell (commander and manual navigation), Fred Haise (LM pilot and life support), and Jack Swigert (CM pilot and power-up). One person can play all roles; two people split the third role.

Choose Mission Control. A referee keeps the answer key hidden, reads each situation aloud, grades committed sheets, and narrates checkpoint consequences. In solo play, cover the answer panel until you commit your sheet; then uncover only that card.

Mission order. Finish cards 11-01 through 11-10 for the beginner game or 13-01 through 13-10 for the advanced game. Launch and intervening coasts are narrated unless a card asks for a calculation. The cards sample the essential game operations; they do not replace hundreds of actual flight-plan procedures.

The role-play loop.

1. The referee announces GET (ground elapsed time since launch), the scene, and the supplied telemetry.
2. The acting pilot reads the problem aloud. Another pilot estimates the order of magnitude and checks units; rotate these jobs each card.
3. Work products, quotients, roots, and logarithms on the slide rule. Use pencil for sums, differences, clock arithmetic, and decisions. Write your scale choices and estimate on the card.
4. Agree on the numerical outputs, signed command, and GO/NO-GO decision. Say "commit" and record them. Corrections before commitment are free; the first committed sheet is the checkpoint.
5. Mission Control grades, applies one reserve charge, and narrates the consequence. Correct an AMBER or RED sheet before continuing.

No hidden arithmetic. All inputs and equations needed for the game appear on the cards or reference sheet. A player's scientific knowledge is not used to add secret hazards, additional fuel charges, or unstated penalties.

Checkpoints: errors have consequences

Command reserve R measures the team's capacity for corrective work, not propellant, hours, or human endurance. Apollo 11 begins at $\boxed{R = 6}$; Apollo 13 at $\boxed{R = 4}$. Reserve is never restored. Physical budgets on the cards remain separate from this abstract track.

Band	First committed sheet	Charge
GREEN (G)	Every output is in its GREEN band and every decision gate is correct. Execute the nominal command.	0
AMBER (A)	At least one AMBER output, no RED output or failed gate. Narrate that card's warning, then correct.	1
RED (R)	Any RED output, wrong sign/quadrant, incompatible or missing units, missing requested output, or wrong discrete decision. Narrate the card's contingency, then correct.	2

Grade the worst result once. Ordinary numerical outputs use

$$\epsilon = 100\,\frac{|q_{\text{player}} - q_{\text{reference}}|}{|q_{\text{reference}}|}.$$

GREEN is $\epsilon \le 1\%$; AMBER is $1\% < \epsilon \le 5\%$; RED is $> 5\%$. Referees compare against guard-digit answers, not the rounded display value. They may use a calculator privately. Three significant figures are normally sufficient.

Clock arithmetic. For h:m:s event times or intervals, GREEN means absolute error at most 1 s, AMBER over 1 s through 60 s, RED over 60 s. These bands do not apply to burn durations computed by division: those use the ordinary relative bands. Decimal hours computed from a clock interval also use ordinary bands.

Angles. For 11-10, 13-08, and 13-09, GREEN is within 0.05°, AMBER over 0.05° through 0.20°, RED beyond. Cartridge / station counts and profile / diversion choices must be exact. Signed negative remainders are valid results when evaluating a rejected profile. If a reference were zero, require the correct zero and units rather than dividing by zero.

Decision gates. A chosen option that violates a stated fuel, oxygen, water, capacity, energy, timing, or entry constraint is RED, even when the individual numerical errors are small. An otherwise correct choice supported by mildly inaccurate numbers is graded AMBER, not automatically RED. Evaluate resource inequalities before rounding. A rounded bound that exactly equals a rounded load is not permission to consume reserve.

Correction protocol. Subtract the charge. If $R \ge 0$, apply the card's narrative and have the crew recompute with a referee hint if needed. Revisions are not charged again. Show a corrected, GREEN-quality pad and a valid gate before advancing. Restore the card's nominal baseline after correction; no numerical state changes occur beyond recorded reserve and narrative flags. This prevents a small slide-rule error from becoming an unplayable cascade. The clock advances to the next card's supplied epoch.

If $R < 0$, stop immediately. Do not solve later cards to declare a rescue. Use the ending rules on the next page.

Mission paths and endings

Apollo 11 / beginner

Launch and Earth parking orbit → TLI (11-01) → docking with Eagle and stage avoidance (11-02) → translunar coast and lunar capture (11-03) → orbit planning (11-04) → undock and DOI (11-05) → powered descent (11-06) → surface EVA (11-07) → ascent and rendezvous (11-08) → LM jettison and TEI (11-09) → entry and recovery (11-10).

Complete all ten corrected gates with $R \geq 0$: **crew recovered; landing achieved**. If any 11-07 warning occurred, add "reduced science" or "EVA curtailed" to the ending. Otherwise record "full planned science." Surviving a warning does not erase it. The final reserve reports command quality, not human survival odds.

If reserve drops below zero on 11-01 or 11-02, record **safe Earth-orbit abort**. On 11-03 through 11-06, record **landing canceled; safe-return contingency ending**. These are narrated training branches, not computed return trajectories. On 11-07 through 11-10, record **simulation terminated; return or rendezvous unresolved**. Do not pretend a landed crew has an automatic rescue.

Apollo 13 / advanced

Narrate launch, docking, and outbound coast. At 055:54:53 the oxygen-tank accident occurs (13-01). Restore free return (13-02), plan oxygen and power (13-03–04), perform PC+2 (13-05), budget water (13-06), adapt CO_2 cartridges (13-07), correct the trajectory manually (13-08–09), and reactivate Odyssey for recovery (13-10).

Complete all ten corrected gates with $R \geq 0$: **all three crew recovered; lunar landing abandoned**. Any reserve exhaustion means **rescue simulation failed; crew survival not established**. Do not invent a successful landing or safe rescue as consolation. The actual Apollo 13 crew returned safely; game failure is not an alternate historical claim.

Campaign play. Finish Apollo 11, then start Apollo 13 with its own fresh reserve of 4. Do not carry reserve or resources between missions. An optional practice session may waive charges, but it is not a scored run.

Resource accounting. Apollo 13 cards 13-03, 13-04, and 13-06 independently cover the same full return horizon from 058:00:00. They are planning audits, not new inventory snapshots taken at each later checkpoint. Do not subtract elapsed-flight consumption again. Cards 13-01, 13-02, 13-05, and 13-10 explicitly use separate pools. Cards 13-07–09 supply their own local scenarios.

The slide-rule flight desk

Operation	Hand procedure
ab, a/b	C/D: align the C index with a on D for multiplication; read D under b on C. For division align b on C with a on D and read under the C index. Choose left or right index to stay on-scale.
a^2, $\sqrt{a}$	D to A gives a square; A to D gives a square root. Choose the A-scale half using the exponent / an order-of-magnitude estimate.
Vector length	Square components on A or C/D, add on paper, then take a square root. Signs disappear only inside the magnitude; retain them on the commanded vector.
10^x, $\log_{10} a$	L reads the logarithmic mantissa associated with D. Keep the characteristic (integer power of ten) on paper. For a small negative exponent, use $10^{-x} = 10^{1-x}/10$.
Trigonometry	T gives $\tan\theta$; slide designs differ, so consult the rule's scale labels. If no T scale, use the table below. Linear interpolation is pencil arithmetic with C/D division.

Decimal places are your responsibility

A slide rule reads significant figures, not powers of ten. Example: 120 m/2 m/s must be near 60 s, not 6 or 600 s. Write the estimate first. Products and quotients require a recorded slide-rule scale operation to count as completed game work. A correct guessed number is not a completed calculation sheet; work it before committing.

Common constants and equations

$$1\text{ ft} = 0.3048\text{ m}, \qquad 1\text{ h} = 60\text{ min} = 3600\text{ s}$$

$$\Delta v = \sqrt{\sum_i \Delta v_i^2}, \quad \bar{a} = \Delta v/t, \quad E = Pt, \quad m = rt, \quad \Delta v = v_e \ln(m_0/m_f).$$

$\ln x = 2.302585 \log_{10} x$. The rocket equation describes an ideal single burn without external forces; use only the card's stated model. Every rate must share the time unit of its duration.

Angle	Tangent	Angle	Tangent
6.0°	0.10510	9.0°	0.15838
6.4°	0.11217	9.5°	0.16734
6.5°	0.11394	10.0°	0.17633
6.8°	0.11924	10.5°	0.18534

Between entries (θ_1, u_1) and (θ_2, u_2), use $\theta \simeq \theta_1 + (u - u_1)(\theta_2 - \theta_1)/(u_2 - u_1)$.

Clock example. GET 102:35:13 means 102 hours after launch, not 10:35 a.m. Add 11:58 as minutes:seconds to get 102:47:11.

Apollo 11: the flight plan becomes a game

The supplied `Apollo11_Flight_Plan.pdf` is NASA's *Final Flight Plan, July 1, 1969*; `flight_plan.txt` is a text rendition of that same Apollo 11 document. The game's chronology follows its planned sequence, not a reconstructed as-flown Apollo 11 timeline.

Planned GET	Event used in play
000:11:43	Earth-orbit insertion; narrated.
002:44:26	TLI; 5 min 20 s.
004:39:44.9	Evasive maneuver after extraction and docking.
075:54:28.4	First lunar-orbit insertion.
080:09:29.7	Second lunar-orbit insertion; narrated.
100:39:50.4	CSM/LM separation; narrated.
101:38:48	Descent-orbit insertion.
102:35:13	Powered descent begins.
102:47:11	Planned touchdown.
124:23:26	Planned ascent.
125:21:19.1	Coelliptic sequence initiation (CSI).
131:53:04.7	Post-rendezvous LM-jettison separation maneuver; narrated.
135:24:33.8	Trans-Earth injection.
195:03:27	Detailed timeline's entry interface.
195:17:22	Detailed timeline's splashdown (EI plus 13 min 55 s).

Start condition. Crew: Armstrong / Aldrin / Collins. Command reserve 6. Launch is successful. Write your reserve on the Apollo 11 ledger before card 11-01.

Why the decimal precision varies. Burn tables provide tenths of a second, while some summary charts round to a minute. The game uses burn tables for ignition/duration and the detailed timeline for entry interface and its elapsed-time table for splashdown. The summary's 195:17 splashdown is rounded to the minute.

What the game does not calculate. The full launch guidance, midcourse orbit determination, landmark tracking, P52 realignment, complete lunar rendezvous, and parachute mechanics are narrated. The ten cards contain all mandatory hand calculations for this game's mission, not all computations made by NASA.

11-01 Commit to the Moon

Timeline / provenance: Planned TLI ignition 002:44:26; 320 s burn. [F1, Table 1-1]

Flight director's situation

Armstrong commands, Aldrin reads the burn pad, and Collins checks the decimal place. The Saturn's third stage will send Columbia and Eagle toward the Moon. Do not mistake feet per second for metres per second.

Your calculation orders. Convert the planned 10,451.2 ft/s velocity change to m/s. Find its average acceleration over 5 min 20 s. Compute the cutoff GET.

Procedure. Convert minutes by hand. On C/D multiply velocity by 0.3048, then divide by elapsed seconds: $\bar{a} = \Delta v / t$. Add the duration to GET on paper. This is a whole-burn average, not constant actual thrust acceleration.

GO / NO-GO gate. Write m/s, m/s^2, and h:m:s on the pad. Missing or incompatible units are RED. The modeled burn is commanded in the forward direction.

If your calculation is wrong. AMBER: guidance flags a targeting discrepancy; charge 1 reserve for a corrective update. RED: the departure targeting test fails; charge 2 for a simulator abort-and-replan. A depleted reserve here means a safe Earth-orbit abort, not a lunar landing.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

11-02 Clear the spent stage

Timeline / provenance: Planned evasive maneuver 004:39:44.9; 2.8 s. [F1, Table 1-1]

Flight director's situation

Collins separates the docked spacecraft from the spent stage. The maneuver pad has two nonzero vector components, not two separate burns. Mission Control wants a magnitude before authorizing ignition.

Your calculation orders. For $(\Delta v_x, \Delta v_y, \Delta v_z) = (5.1, 0, 19.0)$ ft/s, find total velocity change in ft/s and m/s. A training model supplies average acceleration 2.14 m/s^2; find burn duration.

Procedure. Square on A (or multiply C/D), add the squares with pencil, take the square root on A/D: $\Delta v = \sqrt{\Delta v_x^2 + \Delta v_y^2 + \Delta v_z^2}$. Convert on C/D and divide by $\bar{a}$.

GO / NO-GO gate. Use the vector magnitude, not $5.1 + 19.0$. Preserve the supplied component directions on the command pad. The 2.14 m/s^2 acceleration is invented training telemetry.

If your calculation is wrong. AMBER: separation is slower or faster than expected; charge 1 for a tracking update. RED: the safe-separation prediction fails; charge 2 for an emergency avoidance rehearsal before continuing.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

11-03 Capture lunar orbit

Timeline / provenance: Planned LOI1 ignition 075:54:28.4; 358.9 s. [F1, Table 1-1]

Flight director's situation

Columbia passes behind the Moon. Houston cannot coach this burn in real time. Collins reads a retrograde pad; the crew's arithmetic determines whether the simulator accepts lunar capture.

Your calculation orders. Use components $(-2891.8, -433.1, +20.4)$ ft/s to find Δv in ft/s and m/s. Find average acceleration for 5 min 58.9 s. Compute cutoff GET.

Procedure. Use A/D for squares and roots, pencil for addition, C/D for unit conversion and division. Use $\bar{a} = \Delta v / t$. Clock arithmetic is sexagesimal; keep the tenths of a second.

GO / NO-GO gate. Keep all three component signs. A correct magnitude with reversed retrograde direction is RED. This scalar exercise does not compute an orbit or prove capture in the real world.

If your calculation is wrong. AMBER: tracking predicts an off-target lunar orbit; charge 1 for a trim update. RED: the simulator rejects capture; charge 2 for a contingency return rehearsal. Reserve exhaustion means abort the landing and use the game's safe-return ending.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

11-04 Work the lunar clock

Timeline / provenance: Lunar-orbit operations between LOI2 and LM separation. [F1, p. 1-3]

Flight director's situation

Houston schedules a daylight observation and a radio call. Collins must distinguish orbital night from loss of radio coverage: they are different scheduling constraints.

Your calculation orders. An average orbit lasts 118.2 min, including 47 min of night. Find daylight minutes, average angular rate in degrees/min, and the percentage of an orbit with MSFN coverage if coverage is 72 min. Is a 65 min daylight task feasible in one daylight interval?

Procedure. Subtract with pencil, then use C/D for 360/118.2 and 100(72/118.2). Compare the task with daylight duration. Use the supplied full period, not the plan's rounded 3°/min.

GO / NO-GO gate. The 65 min task must fit daylight. Do not treat 72 min of coverage as 72 min of daylight, or claim that their overlap is known. The task duration is a training scenario.

If your calculation is wrong. AMBER: a mistimed observation loses useful coverage; charge 1 for rescheduling. RED: the crew confuses the two windows and misses the handover; charge 2 for a complete scheduling rebuild.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

11-05 Lower Eagle's perilune

Timeline / provenance: Planned DOI 101:38:48, 28.5 s; PDI 102:35:13. [F1, Table 1-2]

Flight director's situation

Aldrin checks descent-orbit insertion while Collins tracks Eagle. Armstrong wants to know how much coast time remains after engine cutoff before powered descent begins.

Your calculation orders. Use the table's required $\Delta v = 70.0$ ft/s. Convert it to m/s; compute average acceleration over 28.5 s. Find DOI cutoff GET and the coast interval from cutoff to PDI.

Procedure. C/D handles conversion and $\Delta v/t$. Add the burn on paper, then subtract cutoff from PDI, borrowing 60 seconds or minutes when necessary.

GO / NO-GO gate. Use required Δv, not a reconstruction from the printed components: this table's DOI components do not exactly reproduce 70. The game does not resolve that source discrepancy.

If your calculation is wrong. AMBER: the coast briefing is late; charge 1 for an expedited pad review. RED: a descent-readiness check is missed; charge 2 for a full rehearsal. If reserve is exhausted, cancel PDI and rendezvous with Columbia in the game's safe-return ending.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

11-06 Land or wave off

Timeline / provenance: Planned PDI 102:35:13 to touchdown 102:47:11. [F1, pp. 1-9, 3-69]

Flight director's situation

Armstrong sees a rough patch ahead. Aldrin calls altitude and fuel. Choose the longest safe diversion before the checkpoint commits the landing. This last-minute hover model is fictional, not Apollo telemetry.

Your calculation orders. First find touchdown GET and average whole-descent acceleration from $\Delta v = 6766$ ft/s and 11 min 58 s. At the terminal gate, altitude is 120 m and downward speed is held at 2.00 m/s. Fuel is 60.0 kg and consumption is 0.700 kg/s. Find descent time, fuel endurance, and maximum extra hover time. Choose the longest safe diversion from 10, 20, or 30 s; find fuel left.

Procedure. Use C/D to convert velocity, divide for average acceleration, divide altitude by speed, and divide fuel by flow. Subtract endurance minus descent time. For a diversion d, $m_{\text{left}} = 60 - 0.700(120/2 + d)$.

GO / NO-GO gate. Choose the longest listed diversion that leaves nonnegative fuel. The terminal constant-speed model is separate from the whole-descent average; do not use that average as net terminal acceleration.

If your calculation is wrong. AMBER: the landing pad needs a last-second correction; charge 1 and land at a less desirable site. RED: fuel prediction rejects the landing; charge 2 and rehearse an immediate wave-off. If reserve is exhausted here, no landing occurs; end with a safe lunar-orbit abort.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

11-07 A walk with a turnaround time

Timeline / provenance: Planned EVA duration 2 h 40 min. [F1, p. 1-2]

Flight director's situation

Armstrong and Aldrin share the science work, but each carries an independent oxygen budget. The flight director will not spend the walk-back reserve on one more rock.

Your calculation orders. For each walker, oxygen starts at 1.80 kg; modeled use is 0.480 kg/h and protected reserve is 0.300 kg. Find use and remaining oxygen over the 160 min EVA. Find maximum allowed EVA duration and schedule margin. Egress and return consume 28 min of the planned EVA; each science station takes 12 min. How many complete stations fit?

Procedure. C/D: convert minutes to hours, multiply by flow, and divide usable oxygen by flow. Pencil: subtract reserves and non-science time. Round station count down, never up.

GO / NO-GO gate. Both walkers must retain at least 0.300 kg. The protected reserve is not available for science. These suit quantities and task times are invented for play.

If your calculation is wrong. AMBER: Houston calls an early turnaround; charge 1 and record "reduced science." RED: the oxygen or station plan violates a return margin; charge 2 and record "EVA curtailed." Correct the budget before ascent preparations.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

11-08 Rise, meet, and dock

Timeline / provenance: Planned ascent 124:23:26, 438 s; CSI 125:21:19.1. [F1, Table 1-2]

Flight director's situation

Eagle leaves the surface. Collins will bring Columbia to the rendezvous only after Aldrin reads back ascent timing and a separate close-range braking command.

Your calculation orders. For ascent $\Delta v = 6060$ ft/s, find m/s and average acceleration over 7 min 18 s. Find insertion GET and time until CSI. At the separate docking exercise, range is 40.0 m and closure speed 0.200 m/s. Find time to contact without braking and a burn duration to cancel closure using 0.0500 m/s^2.

Procedure. Use C/D for conversion and division, pencil for GET. For docking, $t_{\text{contact}} = r/v$ and $t_{\text{brake}} = \Delta v/a$. The nominal burn is initiated immediately; stop the modeled closure before contact and then let the referee narrate docking.

GO / NO-GO gate. Command braking opposite to closure. Wrong direction is RED. The close-range model is not the CSI maneuver and does not model the actual rendezvous orbit.

If your calculation is wrong. AMBER: Collins delays docking while the crew corrects its pad; charge 1. RED: the docking safety predictor flags a collision; charge 2 for an emergency separation rehearsal. Exhaustion after landing means the simulation ends with rendezvous unresolved, not an invented safe rescue.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

11-09 Bring Columbia home

Timeline / provenance: Planned TEI 135:24:33.8, 149.4 s; EI 195:03:27. [F1]

Flight director's situation

After docking and LM jettison, Collins checks the Earth-return pad. Houston supplies signed components and the planned entry-interface time, not a promise that every computed trajectory reaches Earth.

Your calculation orders. For components $(3213.3, 705.0, -138.8)$ ft/s, find vector magnitude in ft/s and m/s, and average acceleration over 2 min 29.4 s. Find cutoff GET and coast time from cutoff to entry interface.

Procedure. Use A/D to square and take a root; pencil to add; C/D to convert and divide. Convert duration before adding GET. Subtract the cutoff time, not ignition, from entry interface.

GO / NO-GO gate. Retain the signed components on the burn command. An incorrect direction is RED even with the right magnitude. Coast time is a schedule calculation, not an orbital prediction.

If your calculation is wrong. AMBER: Earth-return targeting needs a trim; charge 1. RED: the entry predictor rejects the pad; charge 2 for a return-burn replan. Exhaustion means an unresolved return simulation.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

11-10 Entry corridor and recovery

Timeline / provenance: Planned EI 195:03:27; detailed splashdown 13:55 later. [F1, p. 3-135]

Flight director's situation

The crew checks entry geometry before separating for the final descent. Houston's training predictor supplies a downward vertical speed and a horizontal speed. It is a diagnostic exercise, not an atmospheric model.

Your calculation orders. At the training gate, horizontal speed is 11.0 km/s and downward vertical speed is 1.25 km/s. Find flight-path angle $\gamma = -\arctan(v_{\text{down}}/v_{\text{horizontal}})$, speed magnitude, and kinetic energy per unit mass in MJ/kg. Check the modeled corridor $-6.8^\circ \le \gamma \le -6.2^\circ$ and energy limit 63.0 MJ/kg. The detailed plan gives splashdown at EI plus 13 min 55 s. Find splashdown GET and confirm the interval.

Procedure. C/D gives the speed ratio; interpolate the tangent table on the reference sheet or use T. Use A/D for speed. $e = v^2/2$: a squared speed in $(\text{km/s})^2$ gives MJ/kg directly. Use pencil for clock subtraction.

GO / NO-GO gate. Both corridor and energy checks must be GO. Positive γ is an upward trajectory and is RED. These limits are game thresholds, not authenticated Apollo flight rules.

If your calculation is wrong. AMBER: the predictor requests a corrected pad; charge 1. RED: a shallow angle predicts skip-out, a steep angle predicts excessive deceleration, or an energy error flags heating uncertainty; charge 2. Do not enter until corrected. With reserve intact, narrate parachutes and recovery; otherwise the return simulation is unresolved.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

Apollo 13: change the mission, save the crew

Advanced mission / fresh command reserve: 4

Crew: Lovell, Haise, Swigert. Narrate successful launch, docking, and outbound coast. At GET 055:54:53 the service-module oxygen-tank accident occurs. The Moon is now a turning point, not a destination. Your victory condition is recovery of all three crew after ten corrected checkpoints, with command reserve still nonnegative.

Unlike the Apollo 11 campaign, this mission requires independent full-horizon consumable audits, a logarithmic rocket-equation calculation, cartridge-count rounding, signed manual-navigation geometry, and a linear entry-correction model.

Historical anchor	Event used in play
055:54:53	Oxygen-tank rupture; activate the LM lifeboat.
Near 061:30	LM propulsion restores the desired free-return path.
079:27:39	PC+2 begins after the lunar swingby.
Near 105:19	Later manual correction; about a 15 s burn.
Final correction	A 22.4 s burn motivates the entry-predictor exercise.
141:30:02	Aquarius jettison; Odyssey must already be ready.
142:54:41	Historical splashdown and crew recovery.

Planning epoch. Consumable cards 13-03, 13-04, and 13-06 all start their accounting at 058:00:00, regardless of when they are discussed in the narrative. Plan the entire remaining horizon only once per pool. Copy the time horizon to scrap paper and reuse it.

Provenance. The supplied PDF and text are both Apollo 11 resources. NASA's Apollo 13 catalog supplies these rescue event anchors. All game inventories, simplified burn models, and failure thresholds are invented and marked as training data.

Before committing. The commander checks direction, the LM pilot checks the protected reserve, and the CM pilot checks time and units. Any player may catch an error for free before "commit." Start card 13-01 with a fresh ledger; do not carry Apollo 11 resources.

13-01 The alarm: power the lifeboat

Timeline / provenance: Historical tank rupture 055:54:53. [N1]; this activation model is invented.

Flight director's situation

Lovell takes command, Swigert powers down Odyssey, and Haise prepares Aquarius. The explosion occurs regardless of your previous success. Your objective changes from landing to bringing all three crew home.

Your calculation orders. In a separate emergency CM battery pool, usable energy is 2.40 kWh. Load is 1.20 kW and LM activation takes 22 min. Find initial endurance, activation energy, remaining energy, and endurance remaining at that load. Find activation completion GET. Is activation possible before this pool is exhausted?

Procedure. Use C/D: $t = E/P$, $E = Pt$, minutes divided by 60. Subtract energy on paper; add 22 min to the alarm time. This emergency pool is not the protected entry battery on card 13-10.

GO / NO-GO gate. Activate Aquarius immediately and transfer the crew. Keeping Odyssey at this load indefinitely is NO-GO. The explosion is a scripted historical event, never a penalty for bad math.

If your calculation is wrong. AMBER: transition checklist delays threaten battery margin; charge 1. RED: the power-down budget is rejected; charge 2 for a contingency transfer rehearsal. Reserve exhaustion means the rescue simulation fails.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

13-02 Restore a free return

Timeline / provenance: Historical LM return maneuver near GET 061:30; magnitude 11.6 m/s. [N1]

Flight director's situation

The docked stack is not on the desired return path. Haise reads a two-axis training pad while Lovell checks its direction. This burn is distinct from the later PC+2 maneuver.

Your calculation orders. Training components are $(+11.0, +3.70)$ m/s. Find total Δv and burn time using average acceleration 0.400 m/s^2. For a modeled constant thrust 12,000 N, $I_{\text{sp}} = 305$ s, and $g_0 = 9.81$ m/s^2, find mass flow and propellant used.

Procedure. A/D handles vector magnitude. C/D: $t = \Delta v/\bar{a}$, $\dot{m} = F/(I_{\text{sp}} g_0)$, $m_{\text{used}} = \dot{m}t$. Keep intermediate readings on the sheet; do not repeatedly round them. The fixed average acceleration is a supplied approximation.

GO / NO-GO gate. Preserve both positive components. For this isolated maneuver, usable propellant is 150 kg and at least 20 kg must remain. Check the fuel gate; it is not the PC+2 tank inventory.

If your calculation is wrong. AMBER: return tracking predicts a correction; charge 1. RED: a wrong-way or underbudgeted burn fails the free-return predictor; charge 2 for a replanning rehearsal. No lunar landing is available as an alternative ending.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

13-03 Three people, one oxygen budget

Timeline / provenance: Game accounting epoch 058:00:00; historical splashdown 142:54:41. [N1]

Flight director's situation

Aquarius was designed for a different mission. Haise must choose a conservation profile for the entire remaining flight. The same exact time horizon will be used on the electrical and water cards.

Your calculation orders. Compute remaining flight time T in h:m:s and decimal hours. LM usable oxygen is 12.0 kg with a protected 2.00 kg reserve. Compare "conserve" use 0.0350 kg/(person h) against "normal" 0.0500 kg/(person h), for three people. Calculate total use and remaining oxygen for each; choose a safe profile.

Procedure. Subtract GET on paper. On C/D calculate $T = 84 + 54/60 + 41/3600$ and $m = 3rT$. Subtract each use from 12.0 kg. Keep T to at least four digits on paper for later cards.

GO / NO-GO gate. Remaining oxygen must be at least 2.00 kg. Do not confuse per-person use with total cabin use. Carry the chosen profile to the mission ledger; all rates are fictional planning averages.

If your calculation is wrong. AMBER: an incorrect budget prompts tighter rationing; charge 1. RED: the selected profile cannot sustain the protected oxygen margin; charge 2 for an emergency conservation replan.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

13-04 A dark, cold electrical plan

Timeline / provenance: Same 058:00:00–142:54:41 horizon as 13-03.

Flight director's situation

Swigert asks which loads can remain off. Haise proposes a low-power Aquarius profile; Lovell insists on keeping entry energy protected rather than spending it on comfort.

Your calculation orders. Use T from 13-03. LM energy pool is 18.0 kWh; retain 2.50 kWh. Non-idle operations cost 0.600 kWh in addition to idle use. Compare idle loads 0.160 kW (low) and 0.240 kW (normal). Find total use and remaining energy for each, and maximum permissible idle power. Choose a profile.

Procedure. C/D: $E_{\text{use}} = PT + 0.600$. Pencil: $E_{\text{left}} = 18 - E_{\text{use}}$. Then $P_{\max} = (18 - 2.50 - 0.600)/T$. Energy is kWh; power is kW. Do not multiply power by seconds here.

GO / NO-GO gate. The low-power plan must retain at least 2.50 kWh. The 0.600 kWh operations allowance already includes modeled LM burns; do not charge them again. This pool is separate from CM entry energy.

If your calculation is wrong. AMBER: incorrect switching instructions force load shedding; charge 1. RED: the chosen profile predicts depletion before recovery; charge 2 and rewrite the electrical plan. Cold and dim conditions are narrative effects, not additional hidden resource deductions.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

13-05 PC+2: shorten the trip

Timeline / provenance: Historical PC+2 ignition 079:27:39; burn 263.4 s, $\Delta v = 262$ m/s. [N1]

Flight director's situation

After passing the Moon, Aquarius must shorten the return. This is the advanced slide-rule challenge: solve the rocket equation backward, then compare the predicted duration with the historical anchor.

Your calculation orders. For a fictional mass model, $m_0 = 30{,}000$ kg, effective exhaust velocity $v_e = 3000$ m/s, target $\Delta v = 262$ m/s, and average propellant flow 9.525 kg/s. Find final mass, propellant used, and burn duration. Find historical scheduled cutoff using 263.4 s, not your rounded prediction.

Procedure. Use C/D and L: $x = \Delta v/(2.302585 v_e)$, $m_f = m_0\, 10^{-x}$, $m_{\text{used}} = m_0 - m_f$, $t = m_{\text{used}}/\dot{m}$. On L, find the mantissa $1 - x$, read 10^{1-x} on D, then divide by 10 to obtain 10^{-x}. Subtraction uses pencil; retain extra readable digits before subtracting.

GO / NO-GO gate. The isolated burn's usable propellant is 2700 kg; retain 150 kg. Mass must decrease, so $m_f < m_0$. A wrong exponent sign is RED. The historical time is an anchor; the masses and flow are invented.

If your calculation is wrong. AMBER: duration disagrees with the acceleration plan; charge 1 for a throttle-pad correction. RED: the burn budget or exponent is wrong; charge 2 for an emergency targeting rehearsal.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

13-06 Cooling water versus drinking water

Timeline / provenance: Same full return horizon $T = 84.9113889$ h; planning epoch 058:00:00.

Flight director's situation

Haise balances cooling against the crew's drinking allowance. Choose a modeled cooling profile without borrowing the water protected for entry and recovery.

Your calculation orders. Water starts at 100 kg; protect 12.0 kg. Each of three crew uses 0.0600 kg/h for drinking. Compare cooling 0.900 kg/h (normal) with 0.700 kg/h (reduced). Find combined rates, total use, and water left for both profiles. Find the maximum allowable cooling rate, then choose a safe profile.

Procedure. C/D: $r_{\text{total}} = r_{\text{cool}} + 3r_{\text{drink}}$, $m_{\text{use}} = r_{\text{total}}T$. Subtract from 100 on paper. $r_{\text{cool,max}} = (100 - 12)/T - 3r_{\text{drink}}$.

GO / NO-GO gate. Keep at least 12.0 kg. Both proposed cooling settings are assumed thermally permissible in this game; no thermal engineering calculation is implied. The drinking allowance is not optional.

If your calculation is wrong. AMBER: a bad ration sheet prompts an uncomfortable reduced-cooling replan; charge 1. RED: the selected profile consumes protected water; charge 2 for emergency ration planning. Do not invent additional water losses for narrative delays.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

13-07 Make the square cartridge fit

Timeline / provenance: Training checkpoint GET 090:00:00; CO_2 planning to 134:00:00.

Flight director's situation

The astronauts need a cartridge adapter. The referee narrates its construction from available materials; the mathematics determines how many compatible cartridges and how soon they are needed. This is not a real breathing-system construction guide.

Your calculation orders. Three people produce 0.900 kg of CO_2 per person per day. Existing compatible absorbent has 0.600 kg capacity remaining. Each adapted spare removes 1.80 kg. Find hourly production, the load for the next 44 h, and the minimum integer number of spares needed in addition to existing capacity. Find capacity margin, time until existing capacity runs out, and the adapter deadline GET. Assembly takes 2.00 h: find time slack.

Procedure. Use C/D for rate conversion and load. Compute $n = \lceil (m_{\text{load}} - 0.600)/1.80 \rceil$: round UP. Deadline $t = 0.600/r_{\text{hourly}}$. Add deadline to 090:00:00 and subtract assembly time on paper.

GO / NO-GO gate. Finish the adapter before existing capacity expires and allocate enough spares. Timing this card uses decimal endurance bands and clock bands; cartridge count must be the exact minimum integer.

If your calculation is wrong. AMBER: a miscount requires inventory review; charge 1. RED: the predicted scrubber gap would allow CO_2 accumulation; charge 2 for a rushed adapter rehearsal. The referee must not allow the game to proceed with a knowingly inadequate scrubber plan.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

13-08 Navigate without the usual computer

Timeline / provenance: Historical correction near GET 105:19 used about a 15 s burn. [N1]

Flight director's situation

Lovell uses the visible Earth as an attitude reference. Haise works the training pad by hand. There is no die roll for aim: your signed vector and angle decide whether the predictor accepts it.

Your calculation orders. A fictional two-axis correction requires $(\Delta v_x, \Delta v_y) = (-2.30, +0.400)$ m/s. Find magnitude and burn time at average acceleration 0.155 m/s^2. Find angle above the negative x direction: $\theta = \arctan(0.400/2.30)$. Write the commanded quadrant explicitly.

Procedure. Use A/D for the vector root and C/D for division. Use T, or interpolate the tangent table supplied on the reference sheet. Signs define the quadrant; θ is a small positive angle measured from the negative x axis toward positive y, not from positive x.

GO / NO-GO gate. Angle bands are GREEN within 0.05°, AMBER through 0.20°, RED beyond. Wrong quadrant is always RED. Magnitude and time still use ordinary relative-error bands.

If your calculation is wrong. AMBER: the predictor flags aim drift; charge 1 for an attitude reset. RED: a wrong quadrant or angle predicts a missed entry corridor; charge 2 for a complete manual-navigation rehearsal.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

13-09 Thread the entry corridor

Timeline / provenance: Final historical correction lasted 22.4 s; game entry predictor is invented. [N1]

Flight director's situation

Swigert prepares Odyssey while Lovell checks the last correction. The flight director supplies a locally linear response model. It is deliberately simpler than a real Earth-entry trajectory.

Your calculation orders. Current modeled entry angle is $-5.90°$; target is $-6.50°$. A positive corrective Δv changes angle by $-0.0400°$ per m/s. Burn acceleration is 0.670 m/s^2. Find required Δv, nominal burn duration, and predicted angle. Find the minimum and maximum burn durations allowed by $-6.80° \le \gamma \le -6.20°$.

Procedure. Use signed subtraction for $\Delta\gamma$. C/D: $\Delta v = (\gamma_{\text{target}} - \gamma_0)/k$ and $t = \Delta v/a$. Repeat for both corridor edges. Use $\gamma(t) = \gamma_0 + kat$ to check signs.

GO / NO-GO gate. Command a positive correction with a negative angle response. The nominal pad must target $-6.50°$, not merely any duration in the corridor. Angle bands are $0.05°/0.20°$. Report the two duration bounds in ascending order.

If your calculation is wrong. AMBER: a small duration error produces a corridor warning; charge 1. RED: a too-short burn predicts shallow skip-out; too long predicts excessive entry loading. Charge 2 and correct before separation.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

13-10 Wake Odyssey and come home

Timeline / provenance: Historical LM jettison 141:30:02; splashdown 142:54:41. [N1]

Flight director's situation

Swigert powers Odyssey back up. Haise and Lovell check the final energy plan before saying goodbye to Aquarius. You must finish reactivation before jettisoning the lifeboat.

Your calculation orders. A separate, protected CM battery starts with 3.00 kWh. Reactivation uses 0.800 kW for 45 min; entry 1.20 kW for 30 min; recovery beacon 0.150 kW for 4.00 h. Find each energy, total use, and energy remaining; retain 0.500 kWh. Reactivation begins at 140:30:02: find completion GET and margin until LM jettison. Find LM-jettison-to-splashdown interval.

Procedure. Convert minutes to hours on C/D, then use $E = Pt$. Add energy with pencil; subtract from 3.00. Add 45 min to reactivation GET and subtract the historical event times. Beacon use extends after splashdown and is included in this pool.

GO / NO-GO gate. Energy remaining must be at least 0.500 kWh and reactivation must finish before 141:30:02. The CM battery was not charged for the LM's earlier loads; do not merge those inventories.

If your calculation is wrong. AMBER: a sequencing error threatens the transfer; charge 1. RED: premature jettison or insufficient entry energy fails the recovery predictor; charge 2 and correct the sequence. If reserve remains nonnegative and all gates are corrected, narrate parachutes and recovery of all three crew. Otherwise end with rescue unproven.

Calculation record: scales used, estimate, working, units, command.

Committed command / decision: _____

Checkpoint: G / A / R **Reserve after checkpoint:** _____ **Corrected:** yes / no

Apollo 11 / flight director's ledger

Crew names: _____ Date: _____

Initial command reserve: **6**. Charge GREEN 0, AMBER 1, RED 2. Stop if reserve becomes negative. A reserve of exactly zero still permits continuation after correction.

Card	Operation	G/A/R	Charge	R left
11-01	TLI	_____	____	____
11-02	Stage avoidance			
11-03	Lunar capture			
11-04	Orbit windows			
11-05	DOI / coast			
11-06	Landing / diversion			
11-07	EVA / oxygen			
11-08	Ascent / docking			
11-09	Earth departure			
11-10	Entry / recovery			

EVA narrative flag: full science / reduced science / EVA curtailed.

Ending: _____

Crew debrief: What decimal-place, unit, sign, or clock error did your independent check catch before commitment?

Apollo 13 / flight director's ledger

Crew names: _____ Date: _____

Initial command reserve: **4**. Fresh mission, fresh reserve. All consumable figures below are fictional training pools.

Card	Operation	G/A/R	Charge	R left
13-01	Emergency transfer			
13-02	Free-return correction			
13-03	Oxygen plan			
13-04	Electrical plan			
13-05	PC+2			
13-06	Water plan			
13-07	CO_2 adapter			
13-08	Manual navigation			
13-09	Entry correction			
13-10	CM restart / recovery			

Accepted plan	Choice / result	Protected minimum
13-03 oxygen	_____	2.00 kg
13-04 LM energy	_____	2.50 kWh
13-06 water	_____	12.0 kg
13-07 scrubber spares	_____	No capacity gap
13-10 CM energy	_____	0.500 kWh

Accounting horizon: 058:00:00 to 142:54:41. These are independent full-horizon audits, not cumulative deductions. Record only the corrected accepted results.

Ending: _____

Debrief: Which crew member checked the power of ten? Which decision protected a reserve instead of spending it?

Sources, provenance, and modeling boundaries

[F1] Supplied Apollo 11 flight-plan resources. NASA Manned Spacecraft Center, *Apollo 11, AS-506/CSM-107/LM-5, Final Flight Plan*, July 1, 1969. Local files: `Apollo11_Flight_Plan.pdf` (353 PDF pages) and `flight_plan.txt`. The text rendition is not an Apollo 13 flight plan. Printed section page numbers below are preferable to text-file line numbers, which change with extraction format.

Source location	What is used
pp. 1-1–1-3	Phase order; orbit insertion; EVA duration; 118.2 min lunar orbit, 47 min night, 72 min MSFN coverage; summary entry / splashdown schedule.
Figure 1-3, p. 1-7	Powered-descent structure; high/low gates motivate the fictional terminal landing decision.
Table 1-1, p. 1-8	CSM ignition times, burn durations, and supplied vector components for TLI, avoidance, LOI, and TEI. PDF page 30.
Table 1-2, p. 1-9	DOI, PDI, ascent, and CSI timing; required velocity changes. PDF page 31.
Detailed timeline, p. 3-69	PDI 102:35:13, burn 11:58, touchdown 102:47:11.
Detailed timeline, p. 3-135	Entry interface 195:03:27, more precise than the summary's 195:03; splashdown EI plus 13:55, giving 195:17:22.

[N1] Supplemental Apollo 13 history. NASA Space Science Data Coordinated Archive, *Apollo 13 Command and Service Module*, catalog ID 1970-029A, "Mission Profile," accessed October 2, 2026. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1970-029A

This source documents the accident at GET 55:54:53, transfer to the LM, an 11.6 m/s return maneuver, the 263.4 s / 262 m/s PC+2 burn, an approximately 15 s later correction, a 22.4 s final correction, LM jettison, and splashdown at GET 142:54:41. Times given in UTC are converted from launch at April 11, 1970, 19:13:00 UTC: PC+2 April 15 at 02:40:39 becomes 079:27:39; LM jettison April 17 at 16:43:02 becomes 141:30:02. The April 16 correction at 04:32 UTC is near 105:19 GET; the game does not present it as an exact ignition timestamp.

What is invented. Emergency and entry battery pools, resource inventories and rates, allowable reserves, simplified thrust / mass models, docking range and speed, terminal landing fuel and altitude, science-station durations, cartridge capacities, manual-navigation components, entry geometry and linear response, checkpoint bands, command reserve, and every error consequence are game design. No supplied Apollo 13 flight plan was available. Historical Apollo 13 event anchors supplement the Apollo 11 plan's flight-planning structure; its rescue path is not an Apollo 11 timeline with names changed.

Limits. No numerical model here predicts a true orbit, real atmospheric heating, scrubber chemistry, or mission survival. Fixed rates and average acceleration are intentional educational approximations. All real missions required substantial ground and onboard computation; this game does not claim they were flown using only a slide rule.